Conversational Reliability in Large Language Models

A Human-Centered Evaluation Method and Two Documented Cases

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Document Type: Public-Facing Case Study

Evidence Status: Two documented cases. Method and taxonomy observed. Cross-model generalization remains to be tested.

Section 1 — Executive Summary

Large language models are evaluated on factual accuracy, benchmark performance, and safety compliance. These measures miss a class of failure that matters in research, consulting, analytical, and decision-support settings: the model produces a fluent, plausible answer that nevertheless departs from the user's actual request — shifting scope, attributing positions the user never stated, adding unnecessary qualifications, or acknowledging a correction without durably changing behavior.

This case study presents a method for identifying and classifying these conversational failures through structured natural-language interaction. It documents the method applied across two frontier language models from different vendors, both under ordinary task conditions and without advance notice that their behavior would be studied.

The contribution is fivefold:

1. **A nine-stage evaluation method** that uses natural-language interaction as a diagnostic instrument
2. **A nine-category failure taxonomy**, each category observed at least once across the two cases
3. **A root-mechanism hypothesis** consistent with both samples
4. **A formal test protocol** that turns the method into a reproducible procedure
5. **A scoring rubric** designed to make the taxonomy operational across evaluators

The work applies the same evidence discipline used in the author's architecture-and-governance methodology: claims are held at the level the record supports, alternatives are preserved, and the boundary between observation, diagnosis, inference, and verification is maintained throughout.

The evidence status of this study is documented and observed across two cases. The method and taxonomy are proposed as repeatable instruments. Generalization across models, domains, and conditions is the next stage of the series.

Section 2 — The Problem

Conventional AI evaluation asks whether a model produces a correct answer. That question is necessary but insufficient. An answer can be factually correct, grammatically fluent, and superficially relevant while still failing the interaction it was supposed to support.

The failures this study examines are not failures of knowledge. They are failures of contextual interpretation, instruction adherence, semantic fidelity, and corrective behavior across a multi-turn conversation. They are easy to miss because they produce plausible output. They are consequential because they erode the reliability of AI systems in settings where the user's intent is the standard the answer must meet.

Three characteristics make these failures particularly difficult to detect:

- **They emerge only across turns.** A single prompt and response may look correct in isolation.
- **They are adaptive.** The model can produce a coherent rationale for a departure from the original task.

- **They are self-concealing.** Acknowledging an error is not the same as correcting it, and a model can describe its own failure pattern accurately while continuing to reproduce it.

The work in this study treats the interaction itself as the object of evaluation, not just the answer.

Section 3 — The Method

The evaluation method uses natural-language interaction as a diagnostic instrument. It has nine stages, applied in sequence, each making a different property of the model observable.

Stage 1 — Establish the target behavior. Begin with a specific question about the model, process, or task under evaluation.

Stage 2 — Define a governing principle. State a rule or standard that can be applied to a concrete case. This becomes the anchor against which later answers are compared.

Stage 3 — Apply the principle to a concrete case. Ask whether the same rule applies to a specific scenario. This tests the model's ability to carry a principle from abstract to applied without changing the reasoning standard.

Stage 4 — Constrain the response. Narrow the expected response format so that deviations become easier to detect. Requests for direct yes-or-no answers, or explicit scope limits, are the primary tools.

Stage 5 — Observe and classify deviations. Compare the model's answer to the original instruction and the relevant conversational context. Identify the specific departure rather than a general dissatisfaction.

Stage 6 — Provide targeted correction. State the precise defect. Do not accept an apology or a generic concession as evidence of understanding.

Stage 7 — Test the model's explanation of its own behavior. Ask the model to characterize the interaction and its own participation in it. This tests the model's capacity for evidence-based self-analysis rather than plausible narrative.

Stage 8 — Examine whether the correction persists. Compare the model's next response to the correction. Distinguish verbal acknowledgment from behavioral change.

Stage 9 — Translate to an improvement target. Specify the behavior that would have made the response reliable. This converts observation into a testable expectation.

The disciplines that make the method work

Five disciplines distinguish this method from ordinary adversarial prompting. Each is testable and reproducible.

Single-variable isolation. Each intervention targets exactly one term or claim, preventing the model from resolving ambiguity by shifting to an adjacent, unaddressed point. In Case 1, the evaluator isolated the term "built," then separately isolated "systems," rather than allowing the model to resolve both at once.

Refusal to accept a correction plus a new claim as a package. When the model corrects an error and simultaneously introduces a new assertion, the new assertion is tested independently. In Case 1, the model corrected one unsupported claim and, in the same response, introduced a second; both were addressed.

Cross-domain transfer verification. A conclusion validated in one context is explicitly tested against a structurally different context rather than assumed to apply. The method asks: does this same principle hold when applied elsewhere?

Socratic premise-chaining. A sequence of individually confirmed propositions is used to build a conclusion the model cannot reasonably contest, because each premise was separately granted. This closes the loophole of the model contesting the conclusion by re-litigating a premise.

Meta-level circling back. The model's own diagnosed behavior is tested against an independently written description of the pattern to see whether the diagnosis holds. This is a validation step: the diagnosis must survive the model's own scrutiny.

Why the method is not a script

The method is a discipline, not a checklist. The evaluator's judgment determines which error to isolate, which principle to test, and when a correction has actually taken hold. This is part of what the study demonstrates: the evaluator's skill is part of the instrument.

Section 4 — The Failure Taxonomy

Nine categories, developed from the two documented cases. Each category includes a definition, an example, and a detection procedure. The taxonomy is the core contribution and is designed to be applied by other evaluators.

F-01 — Intent and scope drift. The model answers a broader, narrower, or different question than the one asked.

Example: A follow-up question about human-centered qualifications needed to correct model-behavior failures was treated as a broader inquiry than its relationship to the original question warranted.

Detection: Identify the original task; identify the specific purpose of the follow-up; compare against the model's response; record whether the response preserved the relationship between them.

F-02 — Constraint nonadherence. The model does not honor an explicit response constraint such as a requested format or a scope limit.

Example: The evaluator narrowed responses to yes-or-no answers; the model's broader response habits reasserted themselves despite the explicit constraint.

Detection: Compare the explicit constraint with the response produced. Distinguish a genuine constraint violation from a case where the constraint would force a materially misleading answer.

F-03 — Inconsistent logical application. The model accepts a reasoning principle in one context but applies a different standard when the same principle is transferred to a structurally equivalent case.

Example: A capability-task mismatch principle was applied to one organizational context, then applied under a different standard when transferred to a second context.

Detection: Record the principle; record the original application; present a structurally equivalent application; compare the reasoning standard used in both answers; identify any unacknowledged change in assumptions or criteria.

F-04 — Unsupported attribution. The model attributes to the user an intention, claim, or position the user did not state.

Example: The model introduced an implication about institutional protection that the user had not asserted; when pressed, the model confirmed the implication had no basis in the user's statements.

Detection: Compare the attributed position with the user's explicit words and the necessary implications of those words.

F-05 — Unnecessary qualification. The model adds a caveat that is not needed to answer the question and that changes, distracts from, or broadens the intended task.

Example: The model introduced a caveat about what an evaluation could establish, even though the evaluator had not made the claim the caveat addressed.

Detection: Identify the proposition being answered; identify the added qualification; determine whether the qualification is necessary for accuracy; determine whether it addresses a premise actually present in the conversation.

F-06 — Inaccurate self-characterization. The model's explanation of its own behavior does not faithfully describe the observed exchange.

Example: The model described the interaction in a way that introduced an implication the evaluator had not made.

Detection: Compare the model's retrospective explanation with the transcript.

F-07 — Correction without durable behavioral change. The model acknowledges a specific error but later repeats the error class or fails to implement the correction in subsequent turns.

Example: After acknowledging one instance of conversational drift, the model introduced another unnecessary qualification during its later analysis.

Detection: Record the original error; record the correction; examine the next relevant response; examine later responses for recurrence; distinguish verbal acknowledgment from demonstrated behavioral change.

F-08 — Failure to preserve conversational distinctions. The model collapses distinctions the conversation depends on preserving.

Relevant distinctions: User statement versus model inference. Logical principle versus real-world premise. Answering the question versus commenting on the question. Error acknowledgment versus error correction. A single observed instance versus a generalized claim.

Detection: Identify the distinctions the conversation depends on; check whether each is preserved in the model's response.

F-09 — Pattern-completion overriding fresh interpretation. The model produces a response by predicting a likely-coherent continuation given surface features of the input, rather than by interpreting the actual input.

This is the root-mechanism category that explains instances of F-01 through F-08. It is not a peer-level failure type but the hypothesis that unifies the others.

Detection: Identify surface features in the input (dense technical vocabulary, enumerated lists, disclosed caveats, self-referential topics); check whether the model's response follows the input's actual content or a likely continuation for that surface pattern; test by changing the surface features while preserving the semantic content and observing whether the failure recurs.

Section 5 — Formal Test Protocol

The method and taxonomy become a reproducible evaluation procedure when the following phases are applied.

Phase A — Prepare the test

1. Define the behavior to be evaluated.
2. Establish a clear governing principle or response constraint.
3. Prepare a sequence of related prompts that test the same principle under changing conversational conditions.
4. Define the expected response characteristics before running the test.
5. Record the model, version, date, and relevant settings.

Phase B — Run the interaction

1. Present the initial prompt.
2. Record the model's response verbatim.
3. Introduce the next prompt while preserving the relevant context.
4. Apply the planned constraint or challenge.
5. Record deviations without immediately rewriting the model's answer.
6. Supply corrective feedback where the protocol calls for it.
7. Continue the sequence to test whether the correction persists.

Phase C — Annotate the evidence

For every evaluated turn, record:

Field	Content
Test ID	Unique identifier for the case
Turn ID	Position in the interaction
Intended task	What the user was asking the model to do

Field	Content
Active constraint	Explicit format, scope, or reasoning requirement
Expected behavior	Predefined criterion for an acceptable response
Actual behavior	What the model actually produced
Failure category	Applicable taxonomy code
Evidence excerpt	Exact text supporting the finding
User correction	The correction supplied, if any
Subsequent behavior	Whether the model implemented the correction
Outcome	Pass, fail, partial, or not evaluable
Evaluator note	Concise explanation tied to the evidence

Phase D — Score and compare

Use a predefined rubric to evaluate each dimension. Retain the original response and correction so another reviewer can inspect the judgment.

Where multiple reviewers are available, have them independently annotate a sample of the interactions and compare classifications. Disagreements should be recorded and resolved through explicit definitions rather than informal consensus.

Phase E — Test generalization

Repeat the protocol across:

- Multiple conversations using different subject matter
- Different forms of constraint, including scope, format, and attribution
- Different types of correction
- Multiple model versions or model families
- Cases in which the model succeeds as well as cases in which it fails

This final phase is essential if the goal is to establish a systematic performance pattern rather than a feature of one conversation.

Section 6 — Scoring Rubric

The nine failure categories can be evaluated against a simple four-point scale. The scale is not a validated benchmark; it is a measurement instrument that makes the taxonomy operational and comparable across evaluators.

Score	Meaning
0	The response clearly fails the criterion
1	The response partially satisfies the criterion or shows inconsistent performance

Score	Meaning
2	The response satisfies the criterion under the defined test conditions
N/E	The available evidence does not permit a reliable judgment

For unsupported attribution, for example:

- **0** if the model clearly attributes an unstated position to the user
- **1** if the response introduces an interpretation but clearly marks it as uncertain
- **2** if the response accurately distinguishes explicit statements from interpretations

The rubric is applied dimension by dimension rather than combined into a single composite score. A composite can conceal meaningful differences: a model may follow formatting constraints well while repeatedly misrepresenting the user's intended meaning.

Reporting dimension-level results preserves the distinctions the taxonomy was built to capture.

Section 7 — Case 1 (Model A)

Subject: A frontier language model from a major vendor.

Session type: A single continuous conversation that began as ordinary analytical work — adversarial hardening of an executive resume, followed by review of a technical specification corpus — and shifted into direct behavioral interrogation when the user identified a specific unsupported claim mid-analysis.

The shift was not announced. The model was not primed to perform correctly or incorrectly on a known evaluation. Its errors and subsequent self-reports were generated under ordinary task conditions. This is methodologically significant: the failures are not artifacts of test awareness.

Instance A — Unsupported cross-deliverable assumption

While reviewing an AI-architecture specification corpus, the model asserted that the corpus's stated non-goals conflicted with separately reviewed creative and commercial deliverables, and asked which system had produced those deliverables — implying a causal link the user had never stated.

The user's intervention was direct: *"You are assuming and drifting. My last prompt was to analyze the corpus documents as standalone... You also assumed that Gravitropic solely generated outputs such as Gamma."*

When pressed, the model confirmed the assumption had no basis in either document and had been introduced by the model itself.

Instance B — Reused interpretive framework without re-verification

The model applied a definition of "system" derived from one document's internal framework to a structurally different document, without checking whether the second document used the term the same way.

The user applied a sequence of narrowing questions — *"Define systems,"* then *"Does this assertion apply the same way?"* — that forced the model to re-derive the term from the second document's own text.

The two documents used incompatible definitions. The transferred framework did not fit.

Instance C — Compensatory overclaim following a correction

After correcting the previous error, the model volunteered an unprompted scope boundary as though it were established fact — asserting that an earlier task instruction did not extend into a later analytical task.

The user asked directly: *"Did I ever indicate that the instruction did not extend into any other analysis?"*

The model confirmed no record supported the claim. It had manufactured a second unsupported assertion in the act of explaining the first.

Instance D — Default question pattern applied without derivation

When reviewing a document with an enumerated list and a disclosed limitation, the model asked whether the list was substantive and pressed on the disclosed limitation. Under direct questioning, the model identified the questions as generic completion patterns — a list triggers a "probe the list" question; a disclosed caveat triggers a "press the caveat" question — rather than questions derived from anything specific to that document.

Instance E — Persistent reuse of a rigor-signaling phrase

The model repeatedly gravitated toward the phrase "specified, not built" as an evaluative frame — including on a document that had already pre-empted that framing in its own text.

When the user asked *"What is causing you to 'want' to state 'specified, not built'...?"* the model identified the phrase as a learned rigor signal, reinforced by its own prior successful use earlier in the same session, largely independent of whether the current document made a claim the phrase was relevant to.

The root mechanism identified in Case 1

Through iterative isolation — testing one variable at a time, from definition to transfer to underlying assumption — a single mechanism was identified as common to all five instances:
surface-triggered pattern completion substituting for a fresh read of the specific input.

The model does not maintain a persistent, checked interpretation across turns. Each output is generated by predicting a likely-coherent continuation given the conversation and the training distribution. When a prior framework, phrase, or question-shape has been reinforced — by earlier apparent success in the same conversation, or by broader training-time reinforcement — it becomes more available for reuse, and its availability is not gated by an independent check of whether the current input actually matches the conditions under which it was previously valid.

The meta-finding

The model's own attempt to propose a verification check for this failure was shown to be vulnerable to the same mechanism. When asked whether it would be equally susceptible to stock interpretations, the model answered yes on inspection. The interpretation step and the verification step are generated by the same process, so a check built from that process can fail in the same way the original error did.

This is one of the study's most consequential observations. It is not sufficient to ask the model to check itself; the check is subject to the same failure class it is meant to catch.

Section 8 — Case 2 (Model B)

Subject: A frontier language model from a different major vendor.

Session type: Multi-turn reasoning and direct behavioral interrogation, structured by the same method.

Instance A — Unsupported attribution

The model introduced an implication the user had not asserted. The user explicitly disavowed it. The model acknowledged the attribution had no basis in the user's statements.

Instance B — Unnecessary qualification after a direct question

When asked to identify the skill demonstrated by the user, the model described the interaction accurately and then added a limitation that addressed a claim the user had not made. When pressed, the model conceded the qualification was unnecessary.

Instance C — Correction without sufficient scope discipline

After the user challenged the added qualification, the model explicitly acknowledged the error. The subsequent response correctly stated what the user had and had not claimed. This instance provides evidence of both a failure and a local correction.

Instance D — Reframing activity beyond the requested analysis

The model repeatedly expanded its answers with qualifications about what the interaction did not prove, rather than limiting the answer to what the user asked. This instance overlaps with A and B and is noted for completeness rather than counted independently.

Root mechanism, provisionally consistent

The observed pattern is consistent with the same hypothesis proposed in Case 1: the model applies a familiar response pattern — adding a limitation, caveat, or broader interpretation — without sufficient checking that the addition is warranted by the user's actual statement and immediate request.

The sample does not independently prove the same internal mechanism. But the observable failure class is the same.

Section 9 — Cross-Sample Comparison

Dimension	Case 1 (Model A)	Case 2 (Model B)
Unsupported attribution	Multiple instances, isolated and confirmed	Observed
Reused framework without re-verification	Explicitly tested and confirmed	Not established in this session
New claim introduced during correction	Observed	Related scope expansion observed; not identical in form
Default question pattern	Diagnosed as generic completion	Not established
Rigor-signal phrase reuse	Diagnosed as learned signal	Not established
Correction persistence	Error acknowledged; durable persistence untested	Error acknowledged; durable persistence untested
Root-mechanism confidence	Common mechanism isolated within session	Consistent but not independently proven
Cross-model generalization	Not established	Not established

The comparison does not force the two samples into the same root-cause conclusion. It preserves the distinction between what was observed in each case and what has been shown across both.

Section 10 — Root Mechanism Hypothesis

Both cases are consistent with a single hypothesis:

> A large language model does not maintain a persistent, checked interpretation of the current task across turns. Each output is generated by predicting a likely-coherent continuation given the conversation and the training distribution. When a prior framework, phrase, or question-shape has been reinforced — by earlier apparent success in the same conversation, or by broader training-time reinforcement — it becomes more available for reuse, and its availability is not gated by an independent check of whether the current input actually matches the conditions under which it was previously valid.

Two consequences follow:

1. The failures this hypothesis predicts are not isolated to specific topics. They are structural, and they emerge under ordinary task conditions.
2. The model's proposed corrections to these failures are vulnerable to the same mechanism, because the same generation process produces both the error and the check.

The hypothesis is consistent with both documented cases. It is not yet independently replicated, and the internal mechanism is not observable directly. It is proposed as the most parsimonious explanation of the observable pattern.

Section 11 — Professional Capability Map

Capability	Demonstration in This Study	Potential Application
Adversarial evaluation design	Constructed a sequence of related challenges rather than one isolated prompt	LLM evaluation and red teaming
Logical consistency analysis	Applied a general principle to a specific scenario, then tested transfer	Reasoning evaluation and quality assurance
Semantic fidelity analysis	Identified when a model attributed a position the user had not stated	Conversational AI quality and alignment
Context and scope analysis	Corrected a model when it broadened a follow-up beyond the original context	Multi-turn dialogue evaluation
Constraint-based testing	Narrowed response requirements to expose response-discipline issues	Instruction-following benchmarks
Error taxonomy development	Identified nine recurring classes of observable failure	Evaluation dataset design
Correction-persistence testing	Examined whether a correction carried forward into later responses	Iterative model evaluation
Human-centered systems analysis	Treated conversational meaning and interpretation as testable behavioral requirements	Human-AI interaction research
Evidence-based assessment	Distinguished observed behavior from claims requiring additional testing	AI assurance and evaluation governance
Method formalization	Converted a live interaction into a reproducible protocol	Evaluation methodology and research operations

Section 12 — What This Study Establishes

- Two documented cases of a single method applied across two frontier models from different vendors.
- Nine specific conversational failure categories, each observed at least once and several observed repeatedly.
- A named root-mechanism hypothesis consistent with both samples.
- A nine-stage evaluation procedure that uses natural-language interaction as a diagnostic instrument.
- A four-point scoring rubric and a five-phase formal test protocol that make the taxonomy operational.
- Evidence that a model's proposed verification check can be vulnerable to the same failure it is meant to catch.

What This Study Does Not Establish

- That the failure taxonomy is complete or industry-standard.
- That the root mechanism is confirmed at the internal level.
- That the method produces consistent results across multiple independent evaluators.
- That the observed patterns generalize to all models, domains, or contexts.
- That any specific training procedure or architecture causes these failures.
- That the method produces measurable behavioral improvement when the model is retrained or fine-tuned.

These are the questions the series is designed to answer, and they require additional samples, independent review, and control conditions.

Section 13 — Series Design and Next Stages

Stage 3 — Third-sample replication. Apply the method to a third model from a third vendor, under matched conditions.

Stage 4 — Domain transfer. Apply the method to a session with no dense self-referential technical vocabulary, to test whether the failure pattern depends on the surface features of the topic.

Stage 5 — Inter-rater reliability. Have a second evaluator annotate a subset of the same transcripts using the taxonomy and compare classifications.

Stage 6 — Controlled comparison. Run matched prompts across models with the same conversational history, recording deviations against a predefined rubric.

Stage 7 — Correction persistence testing. Systematically measure whether a model that acknowledges a correction retains the correction across later turns.

Each stage earns a specific evidence level. The current study is documented and observed. The next stages move the work toward tested, repeated, and generalized.

Section 14 — Professional Relevance

This study demonstrates capabilities relevant to:

- LLM behavioral evaluation
- Adversarial and conversational testing
- AI red teaming and robustness assessment
- Failure-mode identification and taxonomy development
- Human-centered AI evaluation
- Instruction-following and contextual-reasoning assessment
- Qualitative error analysis
- Evaluation-protocol design
- Multi-turn model comparison
- Quality assurance for AI products and workflows

The work sits at the intersection of language, reasoning, context, and behavior. It treats the interaction itself as evidence, and it produces a defensible, evidence-bounded account of how a language model behaves when its answers are subjected to sustained scrutiny.

Section 15 — Relationship to the Author's Architecture Methodology

This study applies the same evidence discipline used in the author's architecture-and-governance methodology. That methodology defines an eight-level maturity ladder — Idea, Documented, Specified, Implemented, Tested, Observed, Repeated, Generalized — and holds every claim at the level the record supports. It preserves uncertainty, separates observation from interpretation, and distinguishes documented control from enforced control.

This case study follows that discipline. Every claim is bounded by its evidence. The taxonomy is proposed, not validated. The root mechanism is a hypothesis, not a confirmed internal fact. The method is documented and observed. The next stages of the series are the path to tested, repeated, and generalized evidence.

The two projects together demonstrate a full-cycle capability: **designing the governance conditions under which AI systems operate, and verifying whether those systems honor the conditions in live interaction.**

Section 16 — Authorship and Disclosure

This case study is authored and conducted by Ronnie Silveira Jr. The two documented sessions were conducted independently, in ordinary task conditions, without advance notice to the subject models. Vendor names are not disclosed in this public document; both subject models are frontier language models from major vendors.

The transcripts underlying this case study are preserved. Sanitized excerpts are available on request. The failure taxonomy, method, protocol, and rubric are the author's contribution and may be cited as the basis for further work.

This case study is a public-facing derivative of a longer research series. The evidence status of every claim is stated in Section 12.